

Question Number	Answer	Additional guidance	Mark
8(a)(i)	A description that includes two of the following points: • long wings to allow it to {fly long distances / glide / use updraft} (1) • webbed feet to allow it to paddle through water (1) • long beak {to allow it to catch its prey / to feed / for defence} (1)	ACCEPT webbed feet to allow increased air resistance ACCEPT large eyes to see prey in the water	(2)

Question Number	Answer	Additional guidance	Mark
8(a)(ii)	A calculation showing the following steps: • calculation of decrease in number (1) • calculation of population remaining, rounded to a whole number (1)	<u>Example of calculation:</u> 5.3% of 4500 = 238.5 4261 / 4262 Correct answer with no working shown scores full marks	(2)

Question Number	Answer	Additional guidance	Mark
8(b)	An answer that includes the following point: • similar phenotype (1)	ACCEPT they have similar {physical features / anatomy / morphology} ACCEPT could interbreed to produce fertile offspring	(1)

Question Number	Answer	Additional guidance	Mark
8(c)	An answer that makes reference to four of the following: • (genetic) mutation(s) occurred (1) • resulting in (new) allele coding for {larger size / ability to digest meat / eat chicks} (1) • (which then) conferred a selective advantage / (mutated mice) more likely to {survive and reproduce / pass alleles to offspring} (1) • (therefore) increasing allele frequency (1) • mutated mice on this island becoming reproductively isolated (resulting in two different species) (1)	Ignore genes max 3 if they do not refer to {larger size / ability to digest meat / eat chicks / defence from chicks} ACCEPT those without the {mutation / ability to eat chicks / advantageous allele} were less likely to survive and reproduce ACCEPT mice on this island were geographically isolated (from non-mutated mice) / allopatric speciation occurred	(4)

Question Number	Answer	
8 (d)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. <u>On the island</u> • introducing laws to protect the albatross (D) • removal of predators (D) • through trapping / poisoning of mice / introducing a predator of mice (D) • so more chicks will survive to breeding age (E) • fishing exclusion zone (D) • supplying more food to albatross (E) • more food available for parents to give to chicks (E) • therefore increase population size / conserve the albatross (E) • reintroduction program (D) • strategies to ensure increased survival chances of reintroduced birds e.g. removal of mice, behavioural conditioning (E) <u>In zoos</u> • Tristan albatross breeding pairs taken to zoos (D) • captive breeding programmes (D) • reference to studbooks (D) • collecting eggs and taking them elsewhere to hatch (D) • therefore offspring are not eaten by mice / protected from predators (E) • offspring reintroduced to the island (D) • to increase population size (E) • reintroduce Tristan albatross currently held in zoos (D) • therefore increase population size (E) • (captive breeding/studbooks/breeding zoo albatrosses used) to {maintain / increase} genetic diversity (E) • Hardy Weinberg equation used to see change in allele frequency over time (E) (6)	

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Either on island or zoo involvement explained 1 mark = description of one method 2 marks = description of both methods or description of one with some explanation
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Both island and zoo involvement explained in relation to the question 3 marks = description of both methods with some explanation for one 4 marks = description of both methods with explanation
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	All level 2 content plus: consideration of {maintaining / increasing genetic diversity} (Zoo) and / or reintroduction strategies to increase survival (Island)